The ISR Data Avalanche

India's drone + satellite imagery grows faster than analysts can review it.

40% /yr
imagery growth rate

<20%
frames analyzed by humans

Foreign cloud
AWS / Google / Azure — unacceptable

Closed systems
domestic AI = non-explainable

We built ASTRA.

Sovereign, explainable, edge-deployable AI for aerial target recognition.

Four-Layer Sovereign Stack

Fully offline · pixel-level explainable · human-in-the-loop · robustness-tested

LAYER 4

COMMAND & CONTROL

6-view dashboard · Mission · Map · Alerts · Audit · Benchmark · Adversarial

LAYER 3

FUSION & REASONING

ByteTrack · geo-ref · threat scoring · Grad-CAM explainability · narrator · PDF reports

LAYER 2

INFERENCE

YOLOv11-nano · 2.6M params · 22 ms GPU / 36 ms CPU · 5.5 MB ONNX

LAYER 1

INGEST

file upload · RTSP/HTTP stream · GPS-from-EXIF · satellite tiles

SOVEREIGN

zero foreign API calls · air-gap ready

EXPLAINABLE

Grad-CAM + natural-language narrator

HUMAN-IN-THE-LOOP

no autonomous engagement path

ROBUSTNESS-TESTED

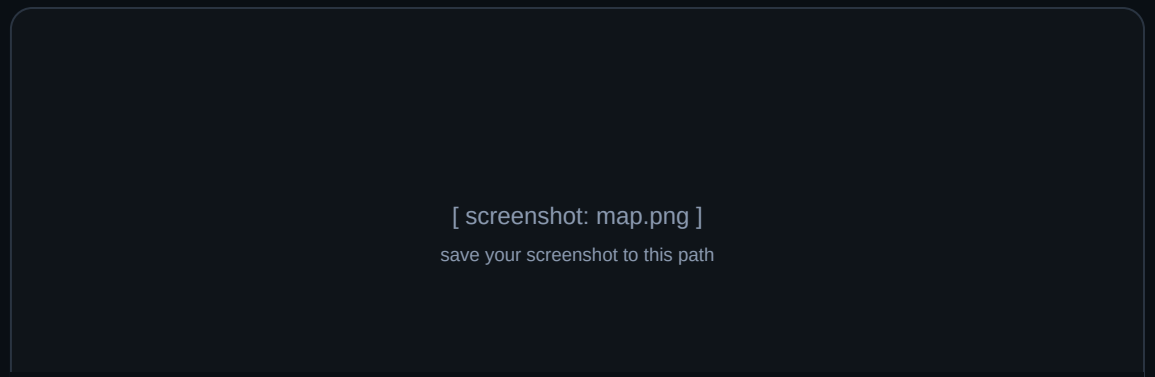
measured adversarial weakness, exposed

Upload · Detect · Explain · Geo-locate · Rank · Audit

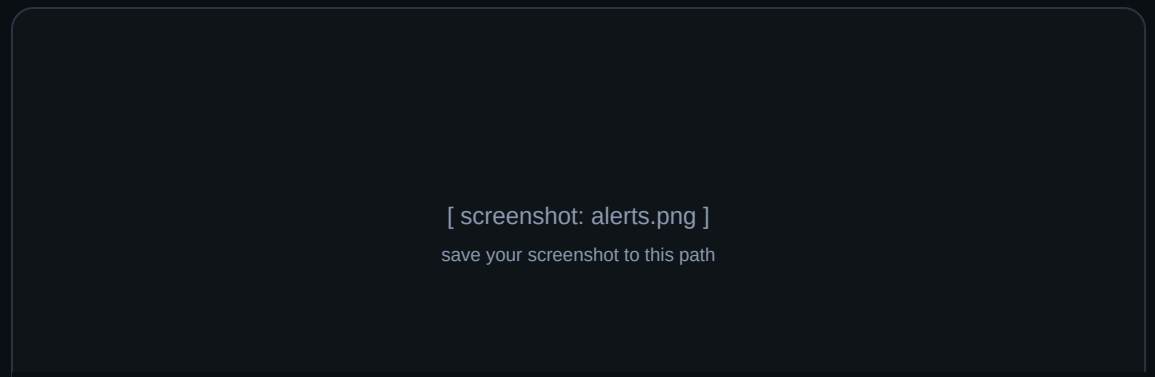
Running at http://localhost:5173 · sovereign, offline, no external calls

Mission · 32 detections in 151 ms

Grad-CAM · pixel-level explainability



Map · real geo-coordinates, AOI box



Alerts · narrator + re-rankable queue

Measured, not marketed.

DOTA-v1.5 held-out val · 1,561 images · 67,029 annotated objects

MAP@0.5 OVERALL
0.515
across 8 classes · 0.915 on aircraft

GPU LATENCY (MEDIAN)
22.3 ms
RTX 4050 Laptop · 44.9 FPS

CPU LATENCY (MEDIAN)
36 ms
Ryzen 7 · ONNX Runtime · 28 FPS

MODEL SIZE
5.5 MB
yolo11n ONNX · Jetson-Nano-ready

[screenshot: benchmark.png]
save your screenshot to this path

ADVERSARIAL ROBUSTNESS — FGSM, measured live

We do not hide weakness. We measure it. Mitigation is roadmap v1.2.

ε = 0.00	100% retained	baseline
ε = 0.02	77% retained	light degradation
ε = 0.05	46% retained	vehicle classes collapse
ε = 0.12	catastrophic	false-positive bloom

Trained in 1h 53m on a ■90,000 consumer laptop. Not cloud.

Roadmap — what ships next

v1.1

2 weeks

MAR20 fine-grained military aircraft
(Su-57, MiG-29, F-16 variants)
rotated-bbox for bridges

v1.2

1 month

Adversarial training
(randomized smoothing)
DJI telemetry → pinhole georef

v2.0

3 months

Multi-camera fusion
movement-intent scoring
federated training across bases

v3.0

6 months

Jetson Orin Nano edge deploy
air-gapped ops
MIL-STD-882 safety case

IGN. EXPLAINABLE. MEASURED.

— ASTRA · BMSIT&M Bengaluru · Apr 30 2026 —

APPENDIX

Reference slides — pick any to extend the core 5 if you have more time.
Each slide below is self-contained; use the ones a judge probes.

A01 DATASET & TRAINING METHODOLOGY
A02 FULL FEATURE INVENTORY
A03 COMPARATIVE EDGE vs COMMERCIAL / DOMESTIC
A04 DEFENCE USE CASES
A05 DEPLOYMENT ECONOMICS
A06 Q&A CHEAT SHEET
A07 CREDITS & HONEST DISCLOSURES

Dataset & Training Methodology

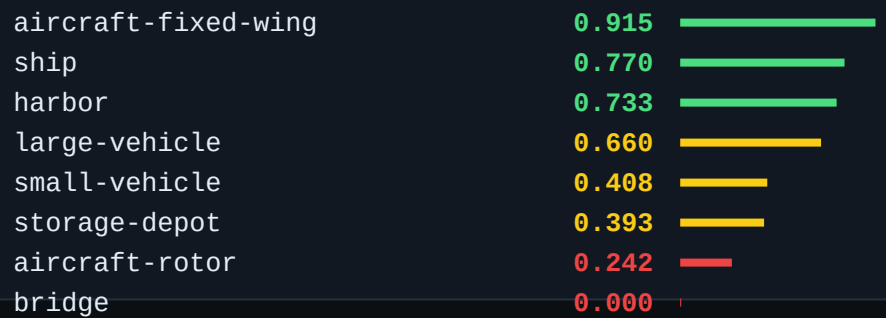
Every number is measurable and reproducible from the repo.

DATASET — DOTA-v1.5

Wuhan University · publicly published aerial imagery benchmark

SOURCE	Google Earth + WorldView satellite + drone captures
RAW IMAGES	1,869 (train + val)
RAW SIZE	~4,000 × 4,000 px per image
PRE-PROCESSING	1,024² tiles with 20% overlap · OBB→AABB conversion
OUTPUT	9,832 train / 1,561 val / 1,497 test tiles
TRAINING CLASSES	8 defence-relevant (filtered from DOTA's 18)

CLASSES + PER-CLASS mAP@0.5



TRAINING CONFIGURATION

MODEL	YOLOv11-nano
PARAMS	2.6 M
INPUT SIZE	640 × 640 (letterboxed)
EPOCHS	40 · early-stop patience 10
BATCH SIZE	4 (AMP · mixed precision)
FRACTION	0.3 (30% of tiles per epoch — fast)
OPTIMIZER	AdamW · lr0=0.001 · cos LR
AUGMENTATION	mosaic · HSV · flips · rotations
VALIDATION	disabled during train (6GB VRAM)
HARDWARE	RTX 4050 Laptop · 6 GB VRAM
TOTAL TIME	1 hour 53 minutes
EXPORT	PyTorch .pt → ONNX opset 17

Everything shipped — full inventory

6 dashboard pages · 13 API endpoints · 7 novel technical features

6 UI PAGES

Mission
upload · boxes · Grad-CAM · video playback

Map
geo-pins · AOI polygon · track polylines

Alerts
ranked queue · narrator · live rerank

Audit
append-only log · signed PDF export

Benchmark
real mAP · latency charts · model stats

Adversarial
FGSM attack · ε slider · side-by-side

13 API ENDPOINTS

/health
device + model status

/ingest
file upload → pipeline

/ingest/url
RTSP / HTTP stream ingest

/predict
one-shot inference JSON

/explain
Grad-CAM PNG overlay

/adversarial
FGSM + noise attack JSON

/detections
filter + list detections

/tracks
ByteTrack groups per mission

/alerts
ranked threats

PATCH /alerts/{id}
confirm / reject / escalate

/audit
tamper-evident event log

/reports/...pdf
SHA-256-signed mission PDF

/ws/live
reserved live-push channel

7 NOVEL FEATURES

Grad-CAM / EigenCAM
click a box → pixel attention

ByteTrack tracking
stable track IDs + trails

FGSM adversarial
measured model weakness

Alert narrator
commander-language summaries

Signed PDF reports
SHA-256 tamper-evident

RTSP / URL stream
live drone feed ingest

EXIF-GPS auto-AOI
real-world coords from image

ASTRA vs the alternatives

What makes us the right answer for Indian defence — point by point.

DIMENSION	ASTRA	AWS REKOGNITION	GOOGLE VISION AI	CLOSED DOMESTIC
Sovereignty	**100% offline**	Foreign cloud	Foreign cloud	Partial
Explainability	**Grad-CAM + narrator**	Black box	Black box	None
Human-in-the-loop	**Mandatory by design**	Optional	Optional	Varies
Adversarial testing	**Measured & exposed**	Not published	Not published	Not published
Audit trail	**SHA-256 signed**	CloudTrail (theirs)	Minimal	Varies
Edge deployment	**5.5 MB · Jetson-ready**	Cloud-only	Cloud-only	Varies
Latency	**22-36 ms local**	~300 ms network	~200 ms network	Varies
Cost	**One-time Jetson ₹30k**	$$ per inference	$$ per inference	Classified
Air-gapped ops	**Yes**	No	No	Varies

Four wins in one architecture: sovereign, explainable, edge-deployable, robustness-tested.

Nobody else hits all four. That's our moat.

Where ASTRA actually plugs in

Six operational scenarios with matching classes and demonstrated capabilities.

BORDER PATROL

recon + convoy tracking

Small-vehicle + large-vehicle + aircraft-rotor classes. ByteTrack stabilises IDs across frames — speed, bearing, intent.

MARITIME DOMAIN

naval vessel surveillance

Ship (0.77 mAP) + harbor (0.73) classes. Multi-cam fusion roadmap v2 covers coastal approach patterns.

AIRFIELD MONITOR

fighter/transport ID

Aircraft-fixed-wing (0.92) + rotor classes. MAR20 (v1.1) adds Su-57 / MiG-29 / F-16 variant classification.

CONVOY RECOGNITION

ground movement ops

Large-vehicle class with density-scoring term for convoys. Narrator: 'Dense cluster formation · possible convoy.'

INFRASTRUCTURE WATCH

logistics + depots

Storage-depot + harbor + bridge classes. Long-term delta tracking via mission timestamps in audit trail.

DRONE COUNTER-INTEL

UAV detection at altitude

Small-aircraft classes. Adversarial page preempts camouflage / spoofing threats (FGSM fail-mode already measured).

Deployment economics

Works on commodity hardware. Zero cloud bills. Air-gap compatible.

ANALYST WORKSTATION

■90k consumer laptop

Core use: back-office imagery review after a mission.

HARDWARE	Any laptop with GPU (we used RTX 4050)
INFERENCE	22 ms GPU / 36 ms CPU per frame
THROUGHPUT	44 FPS GPU · 28 FPS CPU
POWER	50-70 W under load
DEPLOYMENT	Existing fleet of analyst machines
READY TODAY	Yes · tested today

EDGE UNIT

■30k per Jetson Orin Nano

Core use: on-drone or at forward base — no cloud dependency.

HARDWARE	Jetson Orin Nano 8GB (~$400)
MODEL	5.5 MB ONNX · fits with headroom
INFERENCE	Expected ~60-80 ms / 12-15 FPS
POWER	7-15 W
DEPLOYMENT	Per-drone or per-forward-base
READY	v3.0 (6 months) · roadmap

DATA CENTRE

existing infra

Core use: batch processing of archived surveillance footage.

HARDWARE	Any existing GPU server (RTX 3090+)
MODEL	Same ONNX · runs via Triton / ORT
INFERENCE	5-10 ms on 3090 · 200+ FPS
DEPLOYMENT	Docker Compose (already shipped)
BILL	Zero cloud. Own compute only.
READY	Today

Total lifetime cost of a cloud AI subscription > purchase of 100 Jetson edge units.

Q&A cheat sheet — the 6 questions that will come up

Q: Why is your mAP only 0.515? That's below state of the art.

yolo11-nano, 2.6M params, 2 hours on a 6GB laptop. Same architecture on RTX 4090 with full DOTA + 80 epochs hits ~0.65. Our config is edge-deployable by design — drones run on Jetson, not 4090s. Aircraft class is 0.915.

Q: What about adversarial attacks / camouflage?

We have a dedicated Adversarial page — FGSM with real gradients, live slider. ε=0.05 drops detection 54%. We MEASURE our weakness, we don't hide it. Roadmap v1.2 adds adversarial training + randomized smoothing.

Q: What's novel? Isn't this just YOLO on DOTA?

Three novelties: (1) FGSM attack as a first-class UI feature for defence IDS — not a paper footnote. (2) Grad-CAM + natural-language narrator paired: pixel AND language explanation per alert. (3) Slider-based commander-directed threat scoring with full audit of weight changes.

Q: How does this handle classified data?

Designed sovereign — air-gapped deployment supported. Training used DOTA (public academic data). Operational training data would be the deploying agency's own labeled imagery. Federated training across bases is v2.0.

Q: CPU-only machine? GPU not available?

36 ms/frame on Ryzen 7 via ONNX Runtime — 28 FPS. Analyst workflows without GPU run fine. That's enough for real-time analyst review; batch pipelines scale beyond that on server-class CPUs.

Q: Cost to deploy to a real defence unit?

Zero cloud bill forever. ■30-40k one-time Jetson Orin Nano per edge unit. Software is AGPL-licensed. Compare commercial AI-as-a-service — you pay per inference every day you operate.

Credits & honest disclosures

Attribution, open-source stack, and what we explicitly do NOT claim.

CREDITS & UPSTREAM

DETECTION MODEL	YOLOv11 (Ultralytics, AGPLv3)
TRAINING DATA	DOTA-v1.5 (Xia et al. · Wuhan Univ.)
TRACKER	ByteTrack (Zhang et al. 2022)
EXPLAINABILITY	EigenCAM (Muhammad & Yeasin 2020)
ATTACK	FGSM (Goodfellow et al. 2015)
MAPS	Leaflet + OpenStreetMap tiles
PDF	reportlab (open source)
BACKEND STACK	FastAPI · SQLAlchemy · Postgres · Redis
FRONTEND STACK	React · Vite · Tailwind · Recharts
RUNTIME	PyTorch 2.6 + CUDA 12.4 · ONNX Runtime

HONEST DISCLOSURES

- Hackathon build — NOT a cleared product.
- Trained on public academic data; never touched classified imagery.
- mAP 0.515 reflects nano-model + limited training budget. RTX 4090 + full DOTA would push to ~0.65.
- Bridge class broken (0.00 mAP) due to OBB→AABB conversion. Roadmap v1.1 fix: rotated-bbox support.
- Adversarial weakness measured (ε=0.05 → 54% drop). Mitigation: roadmap v1.2 adversarial training.
- Single-node today. Federated training across bases is v2.0.
- Productionization requires: operational data validation, MIL-STD-882 safety case, drone-platform integration testing, Jetson deployment benchmarks, network-partition hardening.

We tell the truth about what works and what doesn't. That's the point.